5V Charger

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Progress Log

Materials

Perf board of size 5cm*7cm, 7805 voltage regulator with a heat sink, 4 silicon diodes, 1000 micro farad capacitor, 100 micro farad capacitor, female USB 2.0 Type-A connector, step down transformer with output of 12V.

Progress Log

–We have been assigned with the project of creating a 5W charger. We went through the pdf file about the charger and saw that there is a mistake in the figure 1.1 (schematic diagram) . Both the terminals of the transformer are connected to the rest of the circuit and the centre tap is connected to the ground. But in the following description of the transformer, it says to connect one terminal to ground and use the centre tap and the other terminal to connect the rest of the circuit. We think that the schematic is wrong because all the calculations are based upon the statement from the paragraph.

–We need to know what each of the components do in the circuit and also find their details. So we began to search about them on the internet and found their functions.

–Instead of building the entire circuit and directly testing it with the mains, we thought it is much safer to build the circuit in a simulation and test it to find if the circuit we have is safe or ends up catching fire.

– We also wanted to see if the circuit given in the pdf has any issues so we ended up verifying it with AI. And the results it gave is that it is not even close to the actual 5W charger. It says to use some safety fuses and and use a MOV, also more capacitors with very less capacitance to filter the current, and it also says that the 100 micro farad capacitor is really low for this project, and it also warned about the extreme heating of the 7805 regulator.

–Now we are confused about how to make it work with the components so we began to search how actual 5W chargers work. They use a component called a buck converter to regulate the

voltage with high efficiency so that they do not heat up considerably. They also use a transformer which works with very high frequency (instead of normal 50Hz of mains) so that the size and weight of the transformer are very less and provide high power.They also use a combination of resistors, capacitors and inductors to make sure that the phone will get a smooth DC charging.

– Since we can't do the project in such a complicated manner, we needed to make a few changes to the given circuit and make it work by pushing the components to their limits.

– We looked for components which may provide a decent performance of the project and eventually decided to have 2 capacitors. A 100 micro farad ceramic capacitor and a 1000 micro farad capacitor. The 1000 farad capacitor will function as the filter for the 7805 and the 100 micro farad ceramic capacitor will function as a smoothing capacitor so that there will be less noise from the high frequency region. Here ceramic capacitor is preferred as they have really less internal resistance so that they can effectively handle the high frequency ripples that can occur at the output.

–We are already running late as other people have already collected the components and even soldered them to test those but we didn't even collect the parts. So we decided to collect the parts and start experimenting with them to see if our circuit works as intended.

–After going to the lab we could not use the ceramic capacitors and ended up using a 100 micro farad electrolytic capacitor for the circuit.

–After returning to the hostel and checking the pdf, we realised that the project is not about a 5W charger but a 5V charger. We felt absolutely stupid. This misunderstanding occurred because whenever we talk about a 5W charger, it basically supplies 5V at a current of 1A and this 5V was the result of the calculations provided in the pdf. The fact that both of us didn't realise this misunderstanding was really terrible.

–Now we have the correct goal of the project and the parts, we need to build the actual circuit and test it.

– The circuit we had before is already perfect for this work so we proceeded into the building phase.

–Since both members of our team don't have any experience with soldering, we needed to be careful to make sure that we don't actually short the circuit and break some parts.

–We tested all the components that we got with a multimeter to see if there are any faulty components and make sure that they are in a good condition.

– When we were testing the resistance of the transformer, we found that it has a fairly high resistance of primary coil and the secondary coil has about 2.5 ohm, which might seem low but

it is high for a secondary coil. Actually it was a safety feature so that by chance there is a short circuit, the internal resistance will limit the current so that there will be less danger. The primary coil resistance is absolutely fine as it has a high number of turns and thinner wire than the secondary. This makes sure that the transformer draws only a very low current from the mains outlet.

–The transformer is actually rated for an output about 0.5-1A current and the 7805 regulator works at its limit while functioning at about 0.5A and all the other parts are also compatible with 0.5A. So we start soldering the components in such a way that the heat sink of the 7805 stays at the edge of the perf board so that when later designing the outer body of the circuit using a 3D printer, the heat sink actually goes outside the outer case.

–We successfully soldered all the components in place and then started testing it. When it is powered on for the first time, we heard some buzzing noise coming from the transformer and we thought that the adhesive holding the laminated plates inside the transformer was a bit damaged but later realised that the solder connecting the input of the transformer is a bit improper. We soldered it again correctly and when turned on later, it did not make any sound.

https://youtube.com/shorts/o8lL_-qRexE

–We used the multimeter to first check the voltage about the usb output but here we made a stupid mistake. The casing of the usb is conducting current when tried to put the probes of the multimeter onto the pins of the usb. Thus it gave fluctuating values. We later realised this and we wanted to check if anything got damaged. Fortunately nothing got damaged and now we tested the voltage across the usb carefully without touching the casing and we were happy to see the expected voltage of 5V. But we also wanted to test the circuit under a load but since we don't have a resistor to use, we had to halt the testing process.

https://youtube.com/shorts/wuIsXtjIEIw

–We started to do the 3D model of the case and it took a few hours to get our desired case. It has 2 separate parts and they are to be held in place by screws. We left plenty of space to make sure that there is no misfit later. But this introduces a new problem of fragile connections with the usb. So we put some extra casing around the usb port. We also added some columns to hold the perf board in place when the whole thing is turned upside down. We made sure that the transformer stays in place when tilted with some more columns. The top part can be removed and all the circuit can be accessed.

–We made another .stl file specifically for the 3D printing. In that file, a few places need some extra support while printing but we are sure that it will print it with not much of a problem.

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